

From Single-Asset Timing to Scenario-Based Portfolio Allocation

A Four-Phase Study of Backtest Design, Parameter Sensitivity, Mean-Variance Allocation, and CVaR

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GitHub Repository: [volatility-aware-backtesting-engine](#)

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Abstract

This report follows four stages in a Python backtesting and portfolio-allocation project. Phase 1 implemented a close-to-close moving-average strategy with a fixed 5% stop and separated signal formation from the next return. Phase 2 replaced the percentage stop with an entry-time Average True Range (ATR) distance. Under parameters selected on AAPL, the change left AAPL's reported 2025 return unchanged, reduced TSLA's return from -5.4% to -6.5% , and raised BTC-USD's return from 8.9% to 13.7% . The result was mixed, so the later phases moved from single-asset timing to cross-asset allocation.

Phase 3 estimated long-only Markowitz portfolios from 2025 daily returns for AAPL, TSLA, BTC-USD, and GLD. Its maximum-Sharpe solution put 89.6% in GLD and had an in-sample Sharpe ratio of 2.65. With the weights fixed, one January-June 2026 holdout returned -2.12% with a Sharpe ratio of 0.01, versus -7.30% and -0.56 for equal weight. Phase 4 fitted Student- t marginals, used a Gaussian copula to simulate 20,000 joint daily returns, and minimized empirical CVaR. The 90% CVaR portfolio returned -3.30% in the same holdout and reduced drawdown and worst-day loss relative to Phase 3, but no portfolio led on every metric.

The cached 2025 sample, the Phase 3 weights, and all Phase 4 scenario allocations were reproduced in a separate numerical audit. The early-2026 prices were downloaded at run time and were not cached, so those holdout figures cannot now be reproduced byte for byte from the repository alone. The study is therefore evidence about implementation choices and risk trade-offs, not persistent alpha.

1 Research Question and Version Design

The project began with an event-order question: how can a backtest prevent a signal calculated at the end of day t from receiving the return already realized during day t ? Once that timing rule was explicit, other problems became more important. A stop selected on one asset did not transfer cleanly to another, and a portfolio optimized on one year looked much weaker in the next window.

Each version therefore changes the question rather than claiming a monotonic improvement.

Table 1: Version sequence and the question addressed at each stage.

Phase	Implementation	Question
V1	MA crossover and fixed 5% stop	Can a close-to-close event loop avoid forward signal leakage, and how does one stop distance behave across assets?
V2	Entry-time $k \times \text{ATR}$ stop	Does scaling the stop by recent price range make the AAPL-selected rule transfer more cleanly to TSLA and BTC-USD?
V3/V3.1	Markowitz allocation and fixed holdout	What changes when the decision variable is portfolio weight, and how stable are 2025 weights in early 2026?
V4	Student- t scenarios and CVaR minimization	How do allocations change when the objective targets simulated tail loss rather than variance or Sharpe ratio?

2 Data and Evaluation Conventions

Phases 1 and 2 downloaded 2025 OHLC data separately for AAPL, TSLA, and BTC-USD. AAPL was used for parameter selection, and the selected parameters were then applied unchanged to the other two assets. Entries and exits were charged a 0.1% commission. Sharpe ratios used 252 annualization days for equities and 365 for BTC-USD. Since the AAPL parameters and AAPL result come from the same year, that result is in sample.

Phases 3 and 4 used the cached file `v3_market_data_2025.csv`. It has 364 dated rows from 1 January to 30 December 2025. Requiring a price for all four assets leaves 249 shared dates and 248 simple daily-return observations. This convention removes BTC weekend observations rather than filling closed equity markets with zero returns. Expected returns were annualized as $252\bar{r}$ and the sample covariance matrix as 252Σ . The risk-free rate was zero.

The fixed holdout ran from 1 January up to, but not including, 15 June 2026. V3.1 and V4 downloaded that window at run time and applied fixed portfolio weights to each day’s asset returns. This is a constant-weight portfolio rebalanced daily without costs. It is one holdout path, not a rolling or repeated validation design.

3 Phase 1: Event Loop and Fixed Percentage Stop

3.1 Signal Timing

For each asset, the strategy formed a Boolean signal from a short and a long moving average. A position opened from the previous close only after the signal changed from false to true. The next daily return was therefore earned after the signal existed.

The stop implementation is deliberately approximate. It checks whether the current close is at or below 95% of the entry price and records an exit at the theoretical stop price. It does not model intraday paths, spreads, or a gap through the stop. The output should be read as a rule-behavior experiment rather than executable performance.

3.2 2025 Results

The AAPL grid selected MA(8, 60). The same pair and the fixed 5% stop then produced Table 2.

Table 2: Phase 1 reported performance under MA(8, 60) and a fixed 5% stop.

Asset	Return	Sharpe	Trades
AAPL	31.6%	2.16	1
TSLA	-5.4%	-0.05	4
BTC-USD	8.9%	0.58	3

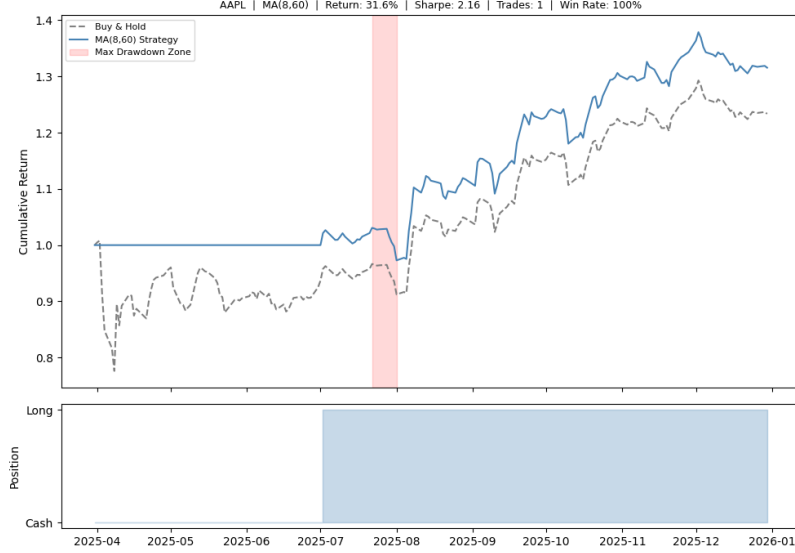


Figure 1: Phase 1 AAPL path under the selected MA(8, 60) rule and fixed stop.

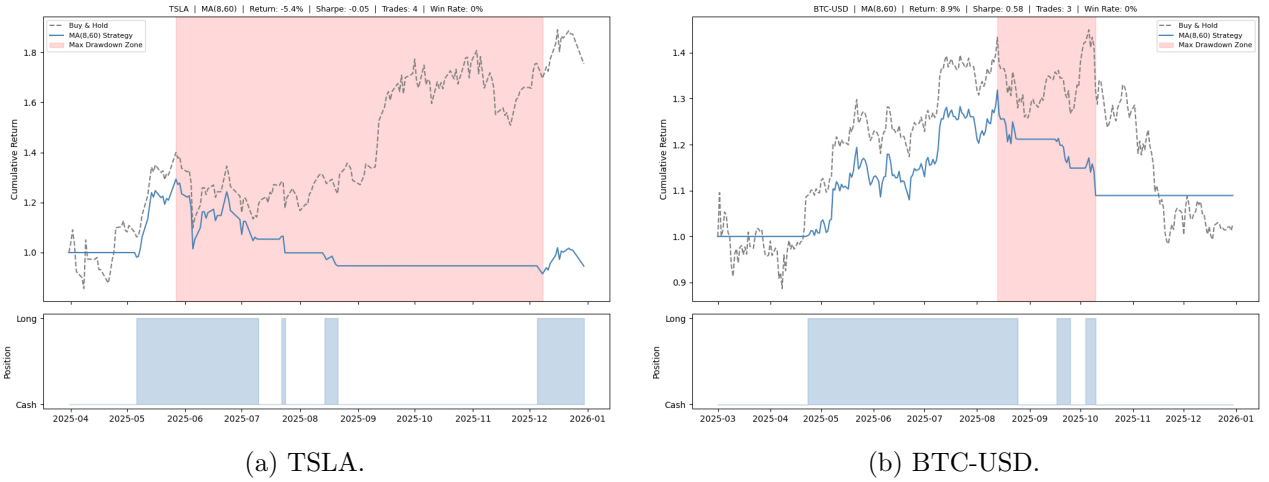


Figure 2: Phase 1 paths under the AAPL-selected rule.

AAPL had one long holding period and finished positive. TSLA moved between cash and long positions and finished slightly negative. BTC-USD remained positive but well below its mid-year peak. The comparison does not isolate the stop from the moving-average signal or the realized path. It does show that a percentage threshold selected on AAPL is not a volatility-neutral control.

4 Phase 2: ATR Scaling and Non-Portable Calibration

Phase 2 replaced the 5% distance with a 14-day Average True Range. For day t ,

$$\text{TR}_t = \max\{H_t - L_t, |H_t - C_{t-1}|, |L_t - C_{t-1}|\},$$

and ATR was the 14-day rolling mean of true range. At entry, the stop was fixed at

$$S_{\text{entry}} = P_{\text{entry}} - k \text{ATR}_{t-1}.$$

This is not a trailing stop. A three-way grid over the short MA, long MA, and k selected MA(8, 60) and $k = 1.5$ on AAPL.

Table 3: Phase 1 and Phase 2 results under their AAPL-selected settings.

Asset	Stop rule	Return	Sharpe	Trades
AAPL	Fixed 5%	31.6%	2.16	1
AAPL	$1.5 \times \text{ATR}$	31.6%	2.16	1
TSLA	Fixed 5%	-5.4%	-0.05	4
TSLA	$1.5 \times \text{ATR}$	-6.5%	-0.09	4
BTC-USD	Fixed 5%	8.9%	0.58	3
BTC-USD	$1.5 \times \text{ATR}$	13.7%	0.84	3

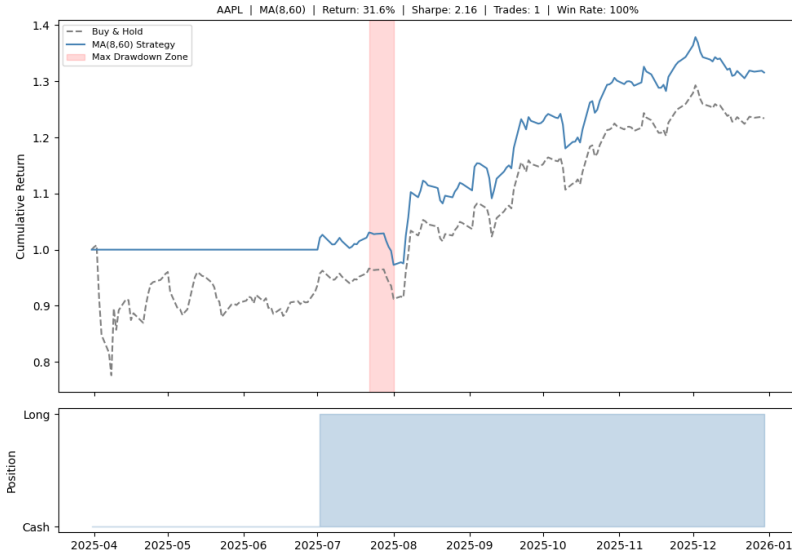


Figure 3: Phase 2 AAPL path under MA(8, 60) and a $1.5 \times \text{ATR}$ entry stop.

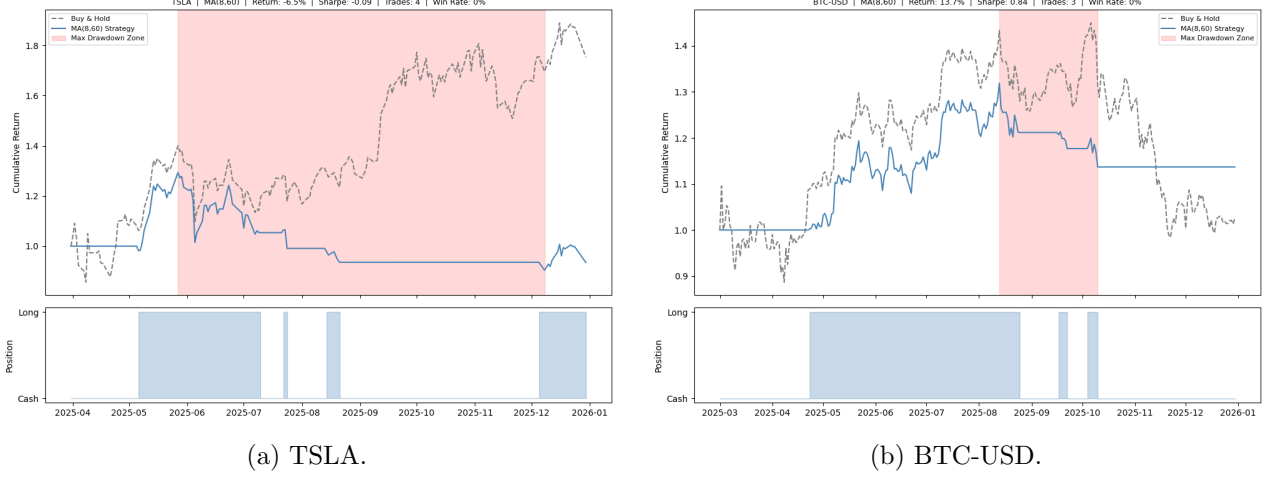


Figure 4: Phase 2 paths under the AAPL-selected rule.

ATR changed the unit of the stop, but it did not make the full rule portable. AAPL was unchanged, TSLA was slightly worse, and BTC-USD improved. Since the moving averages and ATR multiplier were selected together on one AAPL year, the experiment cannot attribute these differences to volatility scaling alone.

5 Phase 3: Mean-Variance Portfolio Allocation

Phase 3 changed the decision from when to hold one asset to how to divide capital across four assets. Let w be long-only weights with $\sum_i w_i = 1$, μ the annualized sample mean vector, and Σ the annualized sample covariance matrix. The two objectives were

$$\min_w \sqrt{w^\top \Sigma w} \quad \text{and} \quad \max_w \frac{w^\top \mu}{\sqrt{w^\top \Sigma w}},$$

corresponding to global minimum variance and maximum Sharpe ratio. Both were solved with SciPy's SLSQP under long-only bounds [1, 3].

5.1 Correlation Structure

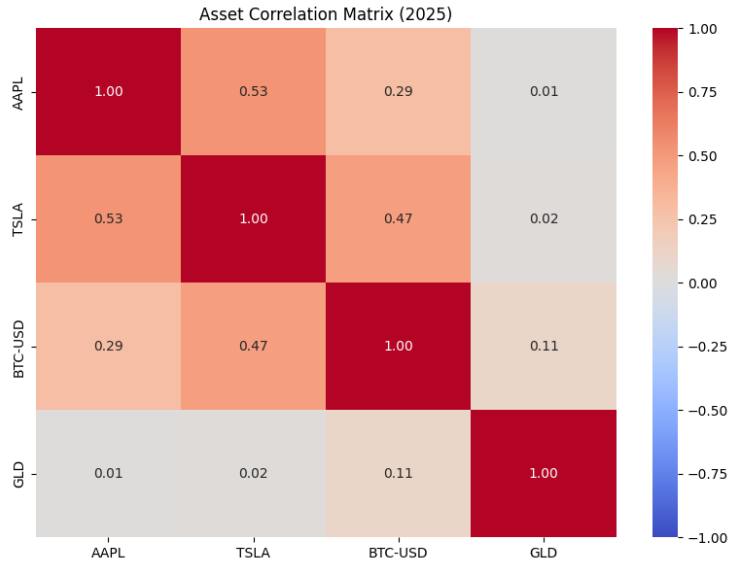


Figure 5: Phase 3 Pearson correlations from 248 daily-return observations on shared 2025 dates.

GLD’s correlations with AAPL, TSLA, and BTC-USD were 0.012, 0.023, and 0.105. These values help explain its large optimized weight. They do not show that the correlations will remain stable outside the training year.

5.2 Optimized Weights and In-Sample Metrics

Table 4: Phase 3 optimized allocations and in-sample annualized metrics.

Portfolio	AAPL	TSLA	BTC	GLD	Return	Volatility	Sharpe
GMV	22.9%	0.0%	8.7%	68.5%	38.98%	16.73%	2.33
Max Sharpe	6.2%	4.3%	0.0%	89.6%	48.71%	18.36%	2.65

The annualized returns are arithmetic scalings of one unusually strong training year, not forecasts. The concentration in GLD is also a warning about estimation sensitivity.

5.3 Random-Portfolio and Multi-Start Checks

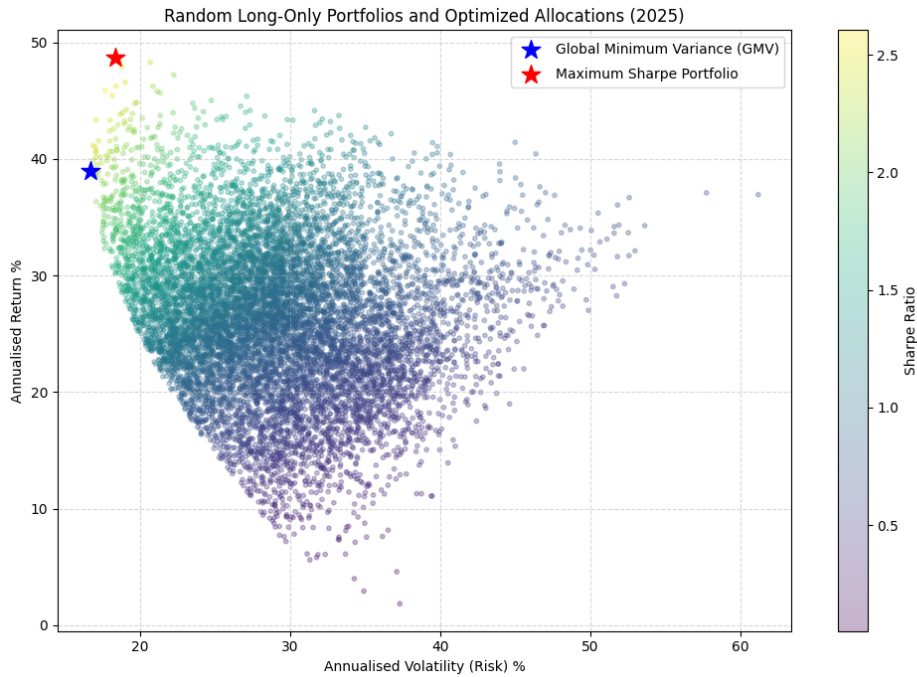


Figure 6: Random feasible portfolios and the Phase 3 optimizer solutions. The cloud is a visual check, not a proof of global optimality.

The optimized points lie on the favorable boundary of 10,000 random feasible portfolios. In the later audit, 50 SLSQP runs from different feasible starting weights converged to the same maximum-Sharpe solution to numerical precision. This is stronger than the plot alone, although it still does not address sampling error in μ or Σ .

5.4 V3.1: Fixed 2026 Holdout

V3.1 froze the full-precision 2025 maximum-Sharpe weights and applied them to the early-2026 window.

Table 5: V3.1 fixed holdout comparison. The configured download interval ends before 15 June 2026.

Portfolio	Cumulative return	Annualized Sharpe
V3 maximum Sharpe	−2.12%	0.01
Equal weight $1/N$	−7.30%	−0.56

The optimized portfolio lost less than equal weight, but its in-sample Sharpe ratio did not persist. One path cannot establish alpha or identify whether the difference came from the mean estimate, covariance estimate, or GLD’s realized return.

6 Phase 4: Student- t Scenarios and CVaR

6.1 Scenario Construction

Phase 4 fitted a univariate Student- t distribution to each asset’s 2025 daily returns. It then used the sample Pearson correlation matrix R as the correlation parameter of a latent Gaussian vector. If

$LL^\top = R$, the simulation was

$$z = L\varepsilon, \quad u_i = \Phi(z_i), \quad \tilde{r}_i = F_{t,i}^{-1}(u_i),$$

where $\varepsilon \sim N(0, I)$, Φ is the standard normal CDF, and $F_{t,i}$ is the fitted marginal for asset i . After the nonlinear marginal transforms, the simulated Pearson correlations need not equal the original sample correlations exactly. The construction also lacks the asymptotic tail dependence of a multivariate t copula.

The code generated 20,000 scenarios with seed 42. If a fitted degrees-of-freedom estimate had been at most two, the marginal would have been refitted with three degrees of freedom. The condition did not activate; the smallest fitted value, for AAPL, was about 2.048.

6.2 CVaR Objective and Numerical Check

For scenario matrix $\tilde{R}$, portfolio loss in scenario s was $\ell_s(w) = -\tilde{R}_s w$. The implemented objective was the mean loss in the worst $1 - \alpha$ fraction:

$$\widehat{\text{CVaR}}_\alpha(w) = \frac{1}{|T_\alpha(w)|} \sum_{s \in T_\alpha(w)} \ell_s(w).$$

This is the empirical tail-loss objective associated with the CVaR optimization framework of Rockafellar and Uryasev [2]. SLSQP reported success at $\alpha \in \{0.90, 0.95, 0.99\}$.

Table 6: Phase 4 optimized scenario CVaR objectives and allocations.

α	Scenario CVaR	AAPL	TSLA	BTC-USD	GLD
0.90	1.7694%	15.29%	0.00%	9.21%	75.50%
0.95	2.2740%	11.80%	0.43%	12.86%	74.91%
0.99	3.5355%	5.36%	2.40%	19.81%	72.42%

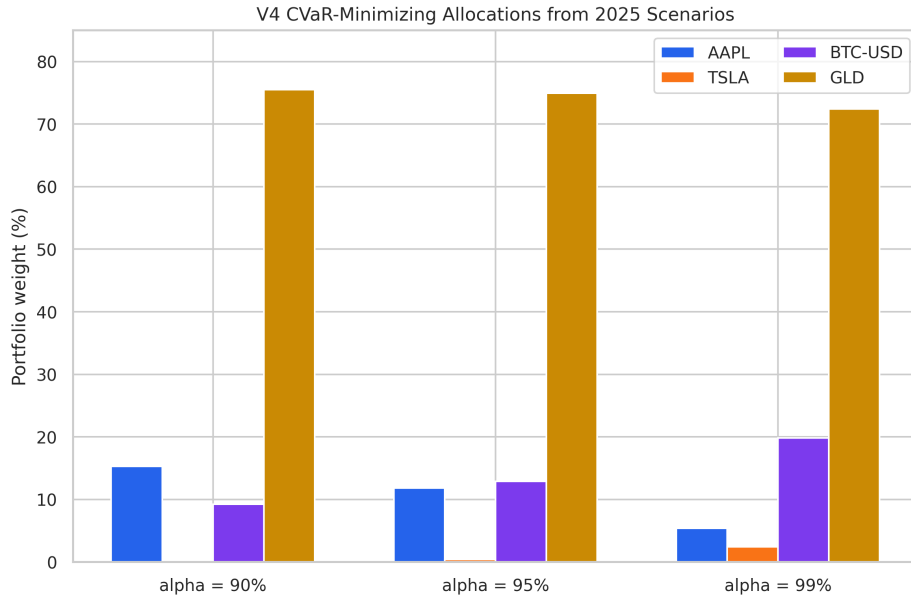


Figure 7: Phase 4 long-only allocations across the three CVaR confidence levels.

A separate audit rewrote the same finite-scenario problem as a Rockafellar-Uryasev linear program and solved it with HiGHS. Relative to the linear-program solutions, the largest SLSQP asset-weight

difference was 0.12, 0.14, and 0.22 percentage points at the three confidence levels. The corresponding objective gaps were below 1.2×10^{-6} . The reported allocations are therefore numerically consistent with the linear-program optimum at the precision used in the report.

The seed still matters because the scenarios are simulated. In eight 20,000-scenario replications, the 90% CVaR AAPL weight ranged from about 11.2% to 17.7%, while GLD ranged from 72.4% to 76.2%. This is simulation uncertainty, not an optimizer failure.

6.3 Fixed 2026 Holdout Comparison

The V4 portfolios were evaluated on the same window as V3.1. The V3 benchmark in this script used four-decimal weights, so it is a close approximation to the full-precision V3.1 portfolio.

Table 7: Fixed 2026 holdout metrics reported by Phase 4.

Portfolio	Return	Sharpe	Max drawdown	Worst day
CVaR 90%	−3.30%	−0.13	−19.41%	−7.73%
CVaR 95%	−4.68%	−0.24	−20.70%	−7.69%
CVaR 99%	−7.30%	−0.43	−22.92%	−7.44%
V3 maximum Sharpe	−2.12%	0.01	−21.66%	−9.03%
Equal weight	−7.30%	−0.56	−15.13%	−4.79%

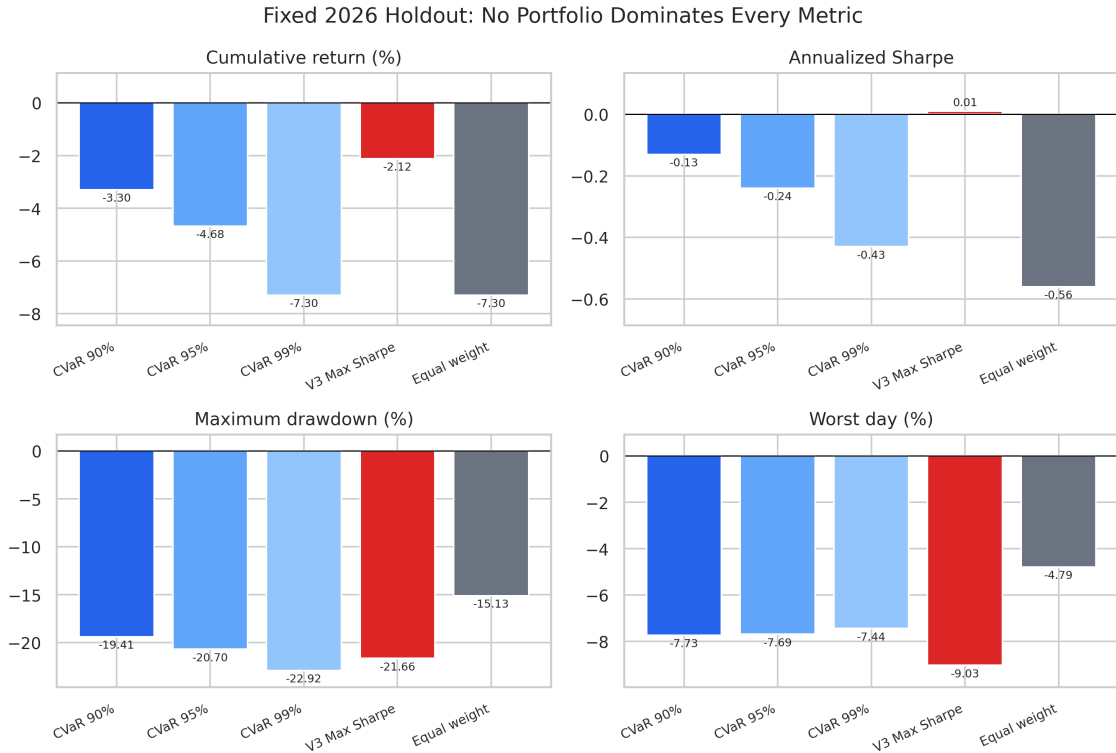


Figure 8: The fixed holdout produced a metric trade-off rather than one dominant portfolio.

V3 had the least negative cumulative return and highest Sharpe ratio. The 90% and 95% CVaR portfolios had smaller drawdowns and milder worst days than V3, while equal weight had the shallowest drawdown and mildest worst day overall. Among the CVaR portfolios, raising α slightly improved the worst day but worsened cumulative return and drawdown. These rankings describe this holdout only.

7 What the Four Phases Establish

1. **Correct event order does not settle the backtest.** Delaying signal use removes one look-ahead error, but fill assumptions and parameter selection still affect the result.
2. **A volatility-scaled unit does not validate a rule.** ATR changes the stop's scale; it does not separate the multiplier, moving averages, and realized path.
3. **The risk objective changes the trade-off.** Mean-variance and scenario CVaR selected different portfolios from the same four-asset sample. None led on return, Sharpe, drawdown, and worst-day loss together.

The V3-to-V4 comparison changes the marginal distributions, dependence construction, and objective at the same time. It cannot attribute the holdout difference to any one component.

8 Limitations and Reproducibility

- **Execution.** V1 and V2 detect a stop breach with the close and fill at the theoretical stop. They omit intraday paths, spreads, slippage, and gap risk. Their trade log also stores the exit-day strategy increment rather than complete trade P&L, so this report does not use the logged win-rate fields.
- **Selection.** V1 and V2 tune on AAPL and report the same AAPL year. V3 and V4 estimate from one calendar year. None of the phases performs rolling or nested model selection.
- **Portfolio mechanics.** V3.1 and V4 apply constant weights to daily returns, which implies costless daily rebalancing. A buy-and-hold portfolio with drifting weights would follow a different path.
- **Distribution and simulation.** Student- t marginals allow heavier univariate tails, but the Gaussian copula has no non-zero asymptotic tail dependence. The latent Gaussian correlation is fixed from the sample, and 20,000 scenarios leave visible seed sensitivity, especially at 99% CVaR.
- **Data versioning.** The cached 2025 prices reproduce the reported Phase 3 and Phase 4 training-sample results. The V1/V2 inputs and early-2026 holdout prices were downloaded live and were not cached. Later revisions or access failures can therefore prevent byte-for-byte reproduction of those figures. During the later audit, the same Yahoo request was rate-limited and returned no usable holdout data.
- **Environment.** The repository does not pin package versions, and the download calls do not set the adjusted-price behavior explicitly. A future rerun may therefore depend on library defaults as well as data revisions.

9 Conclusion

V1 fixed the order in which signals and returns entered the backtest. V2 showed that changing a fixed percentage stop to an ATR-scaled stop did not make the AAPL-selected rule transfer uniformly. V3 moved to portfolio weights, and V3.1 showed how far the in-sample Sharpe ratio fell in one later window. V4 changed the loss function and distributional model; its holdout advantages depended on the metric.

The project does not identify a durable winning portfolio. It does separate three questions that are easy to conflate: whether the code follows its stated event order, whether an optimizer solves its in-sample objective, and whether the resulting allocation performs well later. The cached-data audit supports the first two for V3 and V4. A prospective test with frozen inputs, weights, and evaluation rules is required for the third.

Code and Data Map

File	Role
<code>backtest.py</code>	V1 moving-average engine and fixed stop
<code>backtest_v2_atr_overfit_experiment.py</code>	V2 ATR-scaled entry stop and cross-asset application
<code>backtest_v3_markowitz_portfolio.py</code>	V3 correlation, GMV, maximum Sharpe, and random portfolios
<code>backtest_v3_1_oos_forward_test.py</code>	V3.1 fixed early-2026 holdout
<code>backtest_v4_cvar_scenario_allocation.py</code>	V4 Student- t scenarios, CVaR allocation, and holdout comparison
<code>v3_market_data_2025.csv</code>	Cached 2025 close series used by V3 and V4

References

- [1] H. Markowitz. Portfolio Selection. *The Journal of Finance*, 7(1):77–91, 1952. <https://doi.org/10.1111/j.1540-6261.1952.tb01525.x>.
- [2] R. T. Rockafellar and S. Uryasev. Optimization of Conditional Value-at-Risk. *The Journal of Risk*, 2(3):21–41, 2000. <https://sites.math.washington.edu/~rtr/papers/rtr179-CVaR1.pdf>.
- [3] SciPy Developers. `scipy.optimize.minimize`: SLSQP documentation. <https://docs.scipy.org/doc/scipy/reference/optimize.minimize-slsqp.html>.